SEDEMAC™

MOTOR CONTROLLERS FOR SMALL EVs

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ISAAC 250
Technical Specification

Approximate Product Dimension	
Length	82 mm
Width	49 mm
Height	29 mm
Ingress Protection	IP 67

Key Power Specifications	
Operating voltage range	25V to 42V 36V rated voltage
Continuous motor line current	10A RMS
Peak motor line current for 60 seconds(1)	30A RMS
Rated ambient temperature	45 degC

IO	Example Use Case
5V Sensor Supply	Supply to external sensor assemblies and (optionally) hall sensors
Throttle input	0-5V signal
PAS sensor interface	0-5V signal pulse input
Brake input	0-5V signal
Motor temperature sense	NTC/PTC resistor input
Key switch input	Wake-up / sleep line
Headlamp control	6V / 300 mA high side drive
Taillamp control	(optional) 6V / 300 mA high side drive
CAN H	(optional) CAN high line (non-isolated)
CAN L	(optional) CAN low line (non-isolated)
UART RX	HMI communication
UART TX	HMI communication
Hall A, B, C	(Optional) Hall sensor input for position sensing(2)

(1) Line currents mentioned in "peak motor line current" specification assume that the controller is soaked in ambient temperature of 45 degC before application of peak motor line current.

(2) Hall sensor inputs are not required with SEDEMAC Sensorless Control controller configuration selected.

(3) Reverse polarity protection supported on battery lines.

Note: Product specs are for general reference only and are subject to change without notice. Please check with us on the latest product specifications.

ISAAC 350
Technical Specification

Approximate Product Dimension	
Length	82 mm
Width	49 mm
Height	29 mm
Ingress Protection	IP 67

Key Power Specifications	
Operating voltage range	25V to 42V 36V rated voltage
Continuous motor line current	15A RMS
Peak motor line current for 60 seconds(1)	30A RMS
Rated ambient temperature	45 degC

IO	Example Use Case
5V Sensor Supply	Supply to external sensor assemblies and (optionally) hall sensors
Throttle input	0-5V signal
PAS sensor interface	0-5V signal pulse input
Brake input	0-5V signal
Motor temperature sense	NTC/PTC resistor input
Key switch input	Wake-up / sleep line
Headlamp control	6V / 300 mA high side drive
Taillamp control	(optional) 6V / 300 mA high side drive
CAN H	(optional) CAN high line (non-isolated)
CAN L	(optional) CAN low line (non-isolated)
UART RX	HMI communication
UART TX	HMI communication
Hall A, B, C	(Optional) Hall sensor input for position sensing(2)

(1) Line currents mentioned in "peak motor line current" specification assume that the controller is soaked in ambient temperature of 45 degC before application of peak motor line current.

(2) Hall sensor inputs are not required with SEDEMAC Sensorless Control controller configuration selected.

(3) Reverse polarity protection supported on battery lines.

Note: Product specs are for general reference only and are subject to change without notice. Please check with us on the latest product specifications.

ISAAC 1500
Technical Specification

Approximate Product Dimension	
Length	126 mm
Width	116 mm
Height	59 mm
Ingress Protection	IP 67

Key Power Specifications	
Battery voltage	Intended for vehicles with nominal 48V battery or 60V battery
Continuous motor line current for 30 minutes(1)	47A RMS at 45 degC ambient for 48V system 40A RMS at 45 degC ambient for 60V system
Peak motor line current for 30 seconds(2)	150A RMS at 25 degC ambient for 48V system 135A RMS at 45 degC ambient for 48V system 125A RMS at 25 degC ambient for 60V system 115A RMS at 45 degC ambient for 60V system
Operating temperature limits	-10 degC to +80 degC

IO	Example Use Case
CAN H / CANL	CAN high/low lines (non-isolated)
Digital switches (6 qty)	These are 0-12V digital or 0-48V digital switches. Example use cases are: (a) Brake input (b) Reverse input (c) Mode selection input (d) Multi function switch
SWC / digital switch #7	SWC stands for "single wire communication" which can be used to communicate with HMI. Alternately, the pin can be configured as an additional digital switch.
Ignition key	Wake-up / sleep line
Throttle module (2 qty)	Each throttle module comprises of a 5V signal, a ground return signal, and the throttle signal input
Anti-theft signal	Can be used as an input to anti-theft device to indicate vehicle motion while vehicle is idle.
NTC	Temperature dependent resistor input that can be used to measure motor winding temperature
LSD #1	0.5A / 48V low side drive
LSD #2 / digital switch #8	0.5A / 48V low side drive. Alternately, the pin can be used as an additional digital switch
Position sensor options	(Optional) Digital hall A, B, C interface

(1) Tests conducted with controller inside climate chamber, with nominal air circulation because of thermal management fan of climate chamber.

(2) Test conducted with controller soaked in ambient temperature before application of peak motor line current. After application of mentioned currents, current provided by controller gradually derates and settles at steady state value of ~47A RMS or ~40A RMS depending on controller configuration selected.

(3) Hall sensor inputs are not required with SEDEMAC Sensorless Control.

Note: Product specs are for general reference only, and are subject to change without notice. Please check with us on the latest product specifications.

ISAAC 3500
Technical Specification

Approximate Product Dimension	
Length	146 mm
Width	146 mm
Height	59 mm
Ingress Protection	IP 67

Key Power Specifications	
Battery voltage	Intended for vehicles with nominal 48V battery or 60V battery
Continuous motor line current for 30 minutes(1)	80A RMS at 45 degC ambient for 48V system 70A RMS at 45 degC ambient for 60V system
Peak motor line current for 30 seconds(2)	240A RMS at 25 degC ambient for 48V system 212A RMS at 45 degC ambient for 48V system 200A RMS at 25 degC ambient for 60V system 180A RMS at 45 degC ambient for 60V system
Operating temperature limits	-10 degC to +80 degC

IO	Example Use Case
CAN H / CANL	CAN high/low lines (non-isolated). Sleep / wake-up over CAN is supported.
Digital switches (6 qty)	These are 0-12V digital or 0-48V digital switches. Example use cases are: (a) Brake input (b) Reverse input (c) Mode selection input (d) Multi function switch
SWC / digital switch #7	SWC stands for "single wire communication" which can be used to communicate with HMI. Alternately, the pin can be configured as an additional digital switch.
Ignition key	Wake-up / sleep line
Throttle module (2 qty)	Each throttle module comprises of a 5V signal, a ground return signal, and the throttle signal input
Anti-theft signal	Can be used as an input to anti-theft device to indicate vehicle motion while vehicle is idle.
NTC	Temperature dependent resistor input that can be used to measure motor winding temperature
LSD #1	0.5A / 48V low side drive
LSD #2 / digital switch #8	0.5A / 48V low side drive. Alternately, the pin can be used as an additional digital switch
Position sensor options	Either of below four options can be used: (a) Digital hall A, B, C interface (b) Analog hall U, V, W interface (c) Sin/cos interface (d) SSI interface

(1) Tests conducted with controller inside climate chamber, with nominal air circulation because of thermal management fan of climate chamber.

(2) Test conducted with controller soaked in ambient temperature before application of peak motor line current. After application of mentioned currents, current provided by controller gradually derates and settles at steady state value of ~80A RMS or ~70A RMS depending on controller configuration selected.

(3) Hall sensor inputs are not required with SEDEMAC Sensorless Control.

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ISAAC 4400

Technical Specification

Approximate Product Dimension	
Length	170 mm
Width	150 mm
Height	77 mm
Ingress Protection	IP 67

Key Power Specifications	
Battery voltage	Intended for vehicles with 48V systems
Continuous motor line current with air flow[1]	140 RMS
Peak motor line current for 30 seconds[2]	240A RMS
Rated ambient temperature	45 degC

IO	Example Use Case
CAN H / CANL (2 qty)	CAN high/low isolated CAN lines. One of the lines is intended for actual vehicular application, while the second is intended for development and debugging purposes.
Digital switches (5 qty)	These are 0-12V digital switches. Example use cases are: (a) Brake input - 2 no. (b) Reverse input (c) Mode selection input (d) Multi function switch
12V output	125 mA / 12V output that can be used as supply for digital switches
Pulse train	12V pulse input, for example, to measure wheel speed
Ignition key (2 qty)	Intended to be to KSI and to charger wake-up line
Throttle module (2 qty)	Each throttle module comprises of a 5V/150 mA power supply, a ground return signal, and the throttle signal input
Analog output	0-12V / 200mA analog output. Example use case is to indicate battery SOC to indicator.
NTC	Temperature dependent resistor input that can be used to measure motor winding temperature
LSD (4 qty)	2A / 12V low side drive (1 ON/OFF, 3 PWM)
HSD	2A / 12V high side drive
Position sensor option	Analog hall U, V, W interface. A 5V dedicated regulator is provided.

(1) Airflow of 2 m/s considered.

(2) Line currents mentioned in "peak motor line current" specification assume that the controller is soaked in ambient temperature of 55 degC before application of peak motor line current.

Important Note: 12V and 48V network in the controller are isolated inside the controller. The 12V ground and 48V ground may or may not be shorted in the vehicle outside the controller.

Note: Product specs are for general reference only and are subject to change without notice. Please check with us on the latest product specifications.

ISAAC 15K
Technical Specification

Approximate Product Dimension	
Length	275 mm
Width	232 mm
Height	77 mm
Ingress Protection	IP 65

Key Power Specifications	
Battery voltage	Intended for vehicles with 72V or 96V systems
Continuous motor line current with air flow(1) (3)	250A RMS
Peak motor line current for 30 seconds(2) (3)	550A RMS
Rated ambient temperature	45 degC

IO	Example Use Case
CAN H / CANL (2 qty)	CAN high/low isolated CAN lines. One of the lines is intended for actual vehicular application, while the second is intended for development purposes.
Digital switches (7 qty)	These are 0 -12V digital switches. Example use cases are: (a) Interlock switch (b) Mode switch (c) Sports mode switch (d) Forward switch (e) Reverse switch
Timer capture pin (2 qty)	Feedback, for example, motor fan and MCU fan
Ignition key (1 qty)	Intended to be to KSI
Throttle module (2 qty)	Each throttle module comprises of a 5V signal, a ground return signal, and the throttle signal input
Brake module (2 qty)	Each Brake module comprises the Brake signal input; Uses throttle module 5v signal input and ground return signal
NTC (1 qty)	Temperature dependent resistor input that can be used to measure motor winding temperature
LSD (7 qty)	2A / 12V low side drive (cumulative 10A load)
Position sensor option	(a) Analog hall U, V, W interface. (b) Quadrature AB or ABZ encoder A 5V dedicated regulator is provided.

(1) Airflow of 3.5 m/s considered.

(2) Line currents mentioned in "peak motor line current" specification assume that the controller is soaked in ambient temperature of 45 degC before application of peak motor line current.

(3) External air cooling over the base plate is considered

Important Note: 12V and 72/96V network in the controller are isolated inside the controller. The 12V ground and 72/96V ground may or may not be shorted in the vehicle outside the controller.

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ISAAC Product Line
Feature List

- **Sensorless control: SEDEMAC's flagship offering offered with ISAAC 250, ISAAC 350, ISAAC 1200 and ISAAC 1500**
- **Comprehensive provision for customer to write application software (such as vehicular logics) in Model Based Development workflow with base software package and motor control software provided by SEDEMAC**
- Field oriented control with saddle PWM (i.e., sinusoidal waveforms at line-to-line level with high modulation index).
- Full 4-quadrant operation, which enables optimal regenerative breaking with negative torque control and reverse assist
- Battery current limitation
- Different control modes, such as torque control mode, speed control mode, voltage regulation mode
- Derating functionality as a function of battery voltage, motor temperature, motor currents, I^2T logic etc.
- Rotor flux estimation resulting in magnet strength estimation and diagnostics
- Configurable torque and power limits as a function of motor speed and operating modes, which can enable features such as different assist levels.
- In case of with-sensor control, provision for automatic calibration of the position sensor offsets and motor parameters
- Efficient field weakening logic, that enables higher power extraction at high speeds
- Calibration over CCP
- Diagnostics over UDS / OBD
- Temperature estimator of individual components of controller, such as MOSFETs, capacitors etc. based on a thermal RC model of controller running in real time
- Motor component temperature estimators (stator yolk / winding / magnets etc.) based on thermal RC model of motor running in real time
- Various fault diagnostics, such as line to ground fault, over voltage fault, over temperature fault, over current fault, lock rotor fault etc.

SEDEMAC's vision is to be a **globally respected controls company** developing **innovative technologies and products** that see **widespread use.**

At its core, SEDEMAC is the coming together of talented engineers with the zeal for development and large scale deployment of innovative control technologies & products.

Founded in a lab at the Indian Institute of Technology, Bombay; SEDEMAC currently has world-class development & manufacturing facilities in Pune, India. SEDEMAC's products are now used on **millions** of vehicles and other equipment world-wide.

SEDEMAC sells innovative and critical-to-the-application control products, incorporating fresh control technologies conceived and developed in-house, to leaders of large markets. These products are manufactured in the millions each year in our own world-class manufacturing facilities.

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